

SEO & GEO Auditor - Technical Design

Technical design for an application that audits any website for Search Engine Optimisation (SEO) and Generative Engine Optimisation (GEO), scores it, and produces two separate reports (one for SEO, one for GEO) with recommendations to improve each.

Status: GA4 + Google Search Console use **OAuth CLI** (personal Google account) as the primary auth path. A one-time browser sign-in saves a refresh token; ``backend/connected_properties.json`` maps each domain to its GSC site URL and GA4 Property ID. Up to **6** first-party SEO parameters are scored for connected domains when APIs succeed. Connected Mode also powers a **Performance Baseline** supplementary PDF (23 GA4/GSC + CWV metrics). Service account remains an optional fallback (``GOOGLE_APPLICATION_CREDENTIALS``); OAuth token takes priority when both exist.

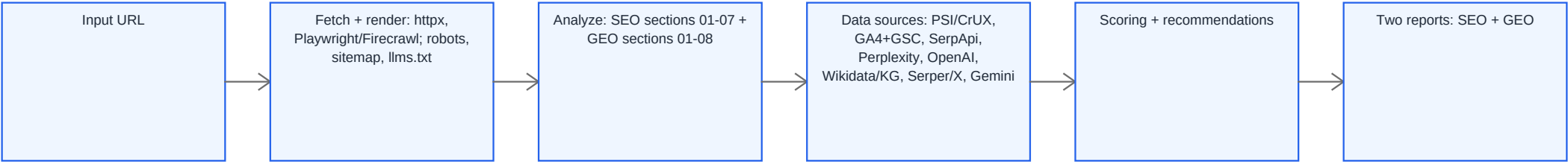
Setup guides (OAuth only, no service account): ``Connected_Mode_Site_Owner_Guide.pdf`` and ``Connected_Mode_Tool_Operator_Guide.pdf`` (regenerate via ``generate_connected_guide_pdf.py``). Regenerate this document's PDF with ``python3 generate_pdf.py SEO_GEO_Auditor_Technical_Design.md SEO_GEO_Auditor_Technical_Design.pdf``.

Recent additions (2026): three Semrush-style supplementary PDFs (Performance Baseline, Full Site Audit, AI Search Overview); unified ``npm run dev`` workflow; SSE audit progress streaming; SQLite audit-history trend charts; PDF parameter comparison tool (``scripts/build_pdf_parameter_comparison.py``).

1. System overview

- A single Python/FastAPI application that exposes a JSON API and also serves a simple React frontend (one deployable unit).
- The user enters a URL; the app crawls/renderers it, runs SEO and GEO analyzers across the full parameter catalog, scores each, and returns two reports.
- Gemini is the LLM used for all AI judgement (content quality, answerability, citation likelihood, etc.).
- SEO is crawl-based; non-crawlable SEO parameters are Manual and excluded from the score, except 6 that are measured when the domain is a connected GA4/GSC property.
- GEO is fully automated using external APIs plus Gemini (no Manual parameters).
- **Supplementary PDFs:** in addition to the in-app SEO/GEO reports, the tool generates three client-facing PDF exports aligned to Semrush reference layouts (Performance Baseline, Full Site Audit, AI Search Overview). These are built with ReportLab from resolved parameter catalogs plus live GA4/GSC bundles when Connected Mode is active.
- **Dev workflow:** ``npm run dev`` from the project root builds the frontend (watch mode) and starts the backend on **``http://127.0.0.1:8000``** (single URL for UI + API).

2. Architecture and data flow



End-to-end: the URL is fetched (raw HTTP + rendered DOM); the SEO and GEO analyzers read the page plus their data sources; results are scored independently; two reports are returned together and shown as two tabs in the UI. Optional shallow multi-page crawl data enriches Full Site Audit and AI Search Overview PDFs. Connected domains trigger parallel GA4/GSC fetches for first-party SEO parameters and for the Performance Baseline PDF analytics bundle.

Backend modules (extended)

Module	Role
<code>`main.py`</code>	FastAPI routes: <code>`/api/audit`</code> , <code>`/api/audit/stream`</code> (SSE progress), <code>`/api/report/pdf`</code> , <code>`/api/report/supplementary-pdf`</code> ; serves built React app
<code>`orchestrator.py`</code>	Concurrent crawl + PSI + agents; <code>`ConnectedRegistry`</code> ; optional <code>`on_progress`</code> callback for streaming

Module	Role
`crawl.py`	Page fetch/render/parse; shallow site crawl graph for supplement agent
`clients.py`	GSC, GA4, PSI, SerpApi, etc.; `Ga4Client.fetch_baseline()` for baseline PDF bundle
`baseline_analytics.py`	Parallel GA4/GSC pull -> `BaselineAnalytics` dataclass for baseline PDF
`baseline_charts.py`	Matplotlib charts embedded in Performance Baseline PDF
`reference_pdf_catalog.py`	Maps Semrush-style PDF parameters -> app parameter keys + manual reasons
`reference_pdf_export.py`	Builds the three supplementary PDFs
`reference_pdf_layout.py`	Shared ReportLab styles, tables, score badges, executive summaries
`pdf_export.py`	SEO/GEO report PDF export
`audit_history.py`	SQLite snapshots for Full Site Audit trend charts (>=2 runs)
`agents/site_audit_supplement_agent.py`	Shallow-crawl issue inventory, site health, internal link distribution
`ai_health.py`	Five-pillar AI Search Health score for AI Overview PDF
`internal_link_distribution.py`	Inlink bucket data for Full Site Audit PDF

3. Tech stack

- Backend: FastAPI + Uvicorn (Python). Serves the API and the built React app.
- Frontend: Vite + React (URL input, live audit progress, SEO/GEO tabs, supplementary PDF download tabs).
- Root tooling: npm + `concurrently` (`npm run dev` runs backend reload + frontend watch build).
- Fetch/render: httpx (raw HTML, headers, redirects) + Playwright/Firecrawl (rendered DOM for JS-heavy pages).
- Parse: BeautifulSoup + lxml.
- Performance: Google PageSpeed Insights API (Lighthouse + lab Core Web Vitals + CrUX field data).
- First-party analytics: google-api-python-client + google-auth + google-auth-oauthlib (OAuth CLI), google-analytics-data (GA4 Data API). Credentials loaded by `app/google\_auth.py` - OAuth refresh token first, service account fallback.
- PDF export: ReportLab (supplementary + SEO/GEO PDFs), Matplotlib (baseline trend/bar charts), pypdf (comparison script text extraction).
- Persistence: SQLite (`backend/data/audit\_history.db`) for Full Site Audit trend history.
- LLM: Gemini via google-genai. OpenAI used only for ChatGPT-style citation checks.

4. Detection methods

- Crawl (auto) = httpx + Playwright/Firecrawl + BeautifulSoup parsing of the page, headers, robots/sitemap/lms.txt.
- PSI = Google PageSpeed Insights API.
- LLM = Gemini judgement.
- External = third-party API (GEO: SerpApi, Perplexity, OpenAI, Wikidata/Wikipedia, Google Knowledge Graph, Serper, X).
- First-party = GA4 / Search Console via OAuth token (or service-account fallback). Only for domains in `connected\_properties.json`. Returns `DetectionMethod.FIRST\_PARTY` with `Confidence.MEASURED` when scored.
- Manual = data the tool cannot obtain without third-party products (off-site link data, GBP) or when the domain is not connected; shown but not scored.
- Not Measured = an automated or first-party check was attempted but could not run (missing/blocked API key, GSC/GA4 API error, PSI unavailable); excluded from the score.

5. Scoring model

- Each scored parameter is rated Meeting (full credit) / Partial (half) / Not Meeting (zero).
- Only Meeting, Partial, and Not Meeting count toward category and final scores. **Manual** and **Not Measured** are shown in the report but excluded from the denominator.

- SEO Manual parameters: **6 off-site** (backlinks, GBP, etc.) are always Manual. **6 first-party** (CTR, sitemap GSC, log file analysis, dwell time, bounce rate, pogo-sticking) are Manual when the domain is not connected; they become First-party when connected and credentials are present.
- GA4 engagement rows (dwell, bounce, pogo) stay **Manual** until GA4 returns session data for the page; GSC rows can still score when GSC responds.
- GEO has no Manual parameters - all are scored or Not Measured.
- Category score = weighted average of that category's scored parameters; final SEO and GEO scores = weighted average across categories, mapped to 0-100 with a letter grade (A-F).

6. Recommendations

- Rule-based: every Partial / Not Meeting parameter carries a concrete fix, a priority (High/Med/Low), and an effort estimate, shown in the report's Recommendation column.
- Gemini-generated: a prioritized narrative summary at the top of each report (top fixes first), generated separately for SEO and GEO.

7. Reports

In-app reports (JSON + UI)

- Two separate reports returned in one API response and shown as two tabs.
- Each report header shows the final score + grade in a coloured circle badge; each category shows a sub-score.
- Each category renders a table: Parameter | What to check | Rating | Recommendation. When a row has both `detail` and `recommendation`, the UI shows the recommendation with evidence `detail` beneath it (muted). SEO Manual and Not Measured rows are visually de-emphasised and noted as excluded from the score.
- SEO report adds a Core Web Vitals panel; GEO report adds an AI-citation appearance panel (which engines cite the site + competitor gap).
- **Audit progress:** `POST /api/audit/stream` emits Server-Sent Events (`progress`, `complete`, `error`) so the UI shows live status ("Generating report...") during long audits.

PDF exports

Export	Endpoint	Description
SEO or GEO report	<code>POST /api/report/pdf</code>	Standard letter-grade report PDF (`pdf_export.py`)
Performance Baseline	<code>POST /api/report/supplementary-pdf` (`kind=performance_baseline`)</code>	Traffic, engagement, GSC/GA4 dashboards, CWV; 23 parameters
Full Site Audit	<code>POST /api/report/supplementary-pdf` (`kind=site_audit_full`)</code>	Site health, errors/warnings/notices, internal links, trend charts; 99 parameters
AI Search Overview	<code>POST /api/report/supplementary-pdf` (`kind=ai_search_overview`)</code>	AI Search Health, bot blocking, crawl status, top issues; 12 parameters

Supplementary PDFs share a common layout system (`reference\_pdf\_layout.py`):

- Branded cover + Heuristics header/footer on audit-style PDFs.
- **Executive summary** (page 2): coloured score badge(s) with label left-aligned, bulleted issues & resolutions (severity colours), optional GA4/GSC KPI lines.
- Compact section spacing (no forced one-parameter-per-page layout).
- Performance Baseline **Overview** KPI grid uses Excel-style full cell borders.
- Full Site Audit shows a single **Site Health** badge in the executive summary (same value as SEO score); issue counts colour red/amber only when count > 0.

- AI Search Overview shows ****AI Search Health**** only (Site Health is scoped to Full Site Audit).

The UI exposes supplementary PDFs as additional download tabs after an audit completes. When Connected Mode is active, ``performance_baseline`` requests trigger ``fetch_baseline_analytics()`` before PDF generation.

PDF parameter comparison

``scripts/build_pdf_parameter_comparison.py`` compares generated PDFs in ``Documents/.../Comparison/`` against Semrush reference PDFs and writes ``PDF-Parameter-Comparison-{date}.pdf`` with per-parameter tables: Reference value | Generated value | Why not calculated (from catalog ``manual_reason``).

Part 1 - SEO Parameters

01 Crawlability & Indexability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
robots.txt	Correct disallow rules; no accidental crawl blocks on key pages	Fetch and parse /robots.txt	Automated	Crawl (no key)
sitemap.xml	Present; no 4xx/5xx URLs; lastmod populated	Fetch sitemap, validate URLs + status	Automated	Crawl (no key)
sitemap GSC-submission status	Submitted to Google Search Console	GSC sitemaps.list (connected); Meeting when submitted with no errors, Partial when errors on submitted sitemaps, Not Meeting when none registered; Not Measured on API error; else Manual	Connected (GSC)	OAuth token (or GOOGLE_APPLICATION_CREDENTIALS fallback)
Meta robots	No unintended noindex/nofollow; check X-Robots-Tag header	Parse meta tag + response headers	Automated	Crawl (no key)
Canonical tags	Self-referencing; no conflicting/chained canonicals	Parse <link rel=canonical>	Automated	Crawl (no key)
HTTP status & redirects	No redirect chains > 1 hop; 301 not 302 for permanent	Follow + record redirect chain	Automated	Crawl (no key)
Crawl depth (NEW)	Key pages reachable within 3 clicks from homepage	Limited internal crawl + depth graph	Automated	Crawl (no key)
Pagination handling (NEW)	rel=next/prev or infinite-scroll pattern; no thin dupes	Parse pagination markup	Automated	Crawl (no key)
Log file analysis (NEW)	Googlebot visit frequency; crawl budget waste	GSC urlInspection.index.inspect (connected); Meeting when indexed and crawled, Partial when crawled but not indexed; Not Measured when empty or API error; else Manual	Connected (GSC)	OAuth token (or GOOGLE_APPLICATION_CREDENTIALS fallback)
XML sitemap index (NEW)	Sitemap index pointing to sub-sitemaps (news/image/video)	Detect <sitemapindex>	Automated	Crawl (no key)
noarchive / nosnippet (NEW)	No accidental suppression of snippets or cached pages	Parse robots directives + headers	Automated	Crawl (no key)

02 On-Page Signals

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Title tag	50-60 chars, primary keyword near front, unique	Parse <title>, length + keyword position	Automated	Crawl (no key)
Meta description	150-160 chars, action-oriented, includes target keyword	Parse meta description	Automated	Crawl (no key)
Heading hierarchy	Single H1; logical H2->H3; no skipped levels	Parse heading tree	Automated	Crawl (no key)
Word count / content depth	Matches competitive benchmark; no thin content	Count words + Gemini depth judgement	Automated	Crawl + GEMINI_API_KEY
Image alt coverage	Informational images have alts; decorative use alt=""	Parse coverage	Automated	Crawl (no key)
Internal links + anchor text	Keyword-rich anchors; no orphan pages; logical graph	Parse links + build internal graph	Automated	Crawl (no key)
Keyword usage	Primary keyword in title/H1/first 100 words; related terms	Text analysis + Gemini semantic check	Automated	Crawl + GEMINI_API_KEY
Duplicate content (NEW)	Near-duplicate pages consolidated via canonical/301	Compare page fingerprints + canonicals	Automated	Crawl (no key)
Content freshness signals (NEW)	Visible publish + last-updated dates	Parse dates in HTML + schema	Automated	Crawl (no key)
Table of contents (NEW)	Anchor-linked ToC on long-form pages	Detect ToC anchors	Automated	Crawl (no key)
E-E-A-T signals on-page (NEW)	Author bio, credentials, About page, attribution	Parse author/about + Gemini judgement	Automated	Crawl + GEMINI_API_KEY

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Outbound link quality (NEW)	Links to authoritative, topically relevant external sources	Parse outbound links + Gemini relevance	Automated	Crawl + GEMINI_API_KEY
Multimedia coverage (NEW)	Video, images, or infographics present	Detect media elements	Automated	Crawl (no key)
LSI / semantic coverage (NEW)	Topic completeness vs top-ranking pages	Gemini topic-model comparison	Automated	GEMINI_API_KEY

03 Technical SEO

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
HTTPS / SSL	Valid cert; no mixed content; HSTS header	Inspect TLS + headers + page resources	Automated	Crawl (no key)
Mobile viewport	Meta viewport; no horizontal scroll; tap targets >= 44px	Parse viewport + PSI mobile audit	Automated	Crawl + PSI (GOOGLE_API_KEY)
URL structure	Lowercase, hyphens, keyword-rich, < 115 chars, no junk params	Analyze URL string	Automated	Crawl (no key)
hreflang	Correct lang/region; self-referencing; x-default set	Parse hreflang tags	Automated	Crawl (no key)
Broken links	0 internal 4xx; external link monitoring	Crawl links + check status codes	Automated	Crawl (no key)
Render-blocking resources	CSS/JS deferred or async; critical CSS inlined	PSI audit + markup parse	Automated	PSI (GOOGLE_API_KEY)
JavaScript rendering (NEW)	Critical content not JS-only	Diff raw HTML vs rendered DOM	Automated	Crawl (Playwright/Firecrawl)
HTTP/2 or HTTP/3 (NEW)	Protocol version check	Inspect connection protocol	Automated	Crawl (no key)
Server response time (NEW)	TTFB < 200ms; server-side caching	Measure TTFB + PSI	Automated	Crawl + PSI (GOOGLE_API_KEY)
Image compression & format (NEW)	WebP/AVIF; lazy loading; explicit width/height	PSI image audit + markup parse	Automated	PSI (GOOGLE_API_KEY)
Third-party script impact (NEW)	Defer non-critical scripts	PSI third-party audit	Automated	PSI (GOOGLE_API_KEY)
Security headers (NEW)	CSP, X-Frame-Options, Referrer-Policy	Inspect response headers	Automated	Crawl (no key)
AMP / Web Stories (NEW)	Valid AMP markup + AMP-canonical pairing	Detect AMP markup	Automated	Crawl (no key)

04 Core Web Vitals & Performance

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
LCP	<= 2.5s; main image/text block is largest element	PSI lab + CrUX field	Automated	PSI (GOOGLE_API_KEY)
CLS	<= 0.1; reserve space for media/embeds	PSI lab + CrUX field	Automated	PSI (GOOGLE_API_KEY)
INP	<= 200ms; replaces FID	PSI / CrUX field	Automated	PSI (GOOGLE_API_KEY)
TTFB	<= 800ms; server + CDN tuning	PSI / measured	Automated	PSI (GOOGLE_API_KEY)
Lighthouse scores	Performance, Accessibility, Best Practices, SEO >= 90	PSI category scores	Automated	PSI (GOOGLE_API_KEY)
Field data (CrUX) (NEW)	Real-user field data; lab != real users	PSI CrUX data	Automated	PSI (GOOGLE_API_KEY)
Mobile vs desktop delta (NEW)	Run CWV separately for mobile and desktop	PSI run twice	Automated	PSI (GOOGLE_API_KEY)
Font loading strategy (NEW)	font-display: swap; preload key fonts; fallbacks	Parse CSS/font usage + PSI	Automated	Crawl + PSI (GOOGLE_API_KEY)

05 Structured Data & Social

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Schema.org JSON-LD	Article, Product, FAQ, HowTo, BreadcrumbList, Organization, LocalBusiness	Parse + validate JSON-LD	Automated	Crawl (no key)
OG / Twitter cards	og:title, og:image (1200x630), og:description, twitter:card	Parse meta tags	Automated	Crawl (no key)
Favicon	SVG + 32px PNG + apple-touch-icon	Detect favicon links	Automated	Crawl (no key)
Rich result eligibility (NEW)	Warnings/errors check	Validate schema against rich-result rules	Automated	Crawl (no key)
Breadcrumb schema (NEW)	BreadcrumbList on interior pages	Parse JSON-LD	Automated	Crawl (no key)
Sitelinks searchbox (NEW)	WebSite schema with potentialAction SearchAction	Parse JSON-LD	Automated	Crawl (no key)

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Review / Rating schema (NEW)	AggregateRating on product/service pages	Parse JSON-LD	Automated	Crawl (no key)
Video schema (NEW)	VideoObject with thumbnail, duration, uploadDate	Parse JSON-LD	Automated	Crawl (no key)

06 Off-Page & Authority

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Backlink profile	Domain authority distribution; toxic ratio; anchor diversity	Requires off-site link index	Manual	Requires off-site link data
Referring domain count	Unique linking root domains; trend	Requires off-site link index	Manual	Requires off-site link data
Topical authority	Topic cluster breadth; hub-spoke internal linking	Internal graph + Gemini topic analysis	Automated	Crawl + GEMINI_API_KEY
Brand mentions (unlinked)	Citation signals across the web	Requires off-site monitoring	Manual	Requires off-site monitoring
Competitor link gap	Links competitors have that you lack	Requires off-site link index	Manual	Requires off-site link data
NAP consistency (local) - on-page	Name/Address/Phone present and consistent on page	Parse on-page NAP	Automated	Crawl (no key)
NAP consistency (local) - cross-directory	Identical across external directories	Requires directory data	Manual	Requires off-site data
Google Business Profile	Verified, complete listing; reviews; posts	Requires GBP ownership/listing data	Manual	Requires GBP

07 UX & Engagement Signals

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Click-through rate (CTR)	GSC CTR vs position benchmark	GSC searchanalytics.query (connected); scored Not Meeting when 0 impressions; Not Measured on API error; else Manual	Connected (GSC)	OAuth token (or GOOGLE_APPLICATION_CREDENTIALS fallback)
Dwell time / scroll depth	GA4 engagement proxy	GA4 runReport (connected + data), else Manual	Connected (GA4)	OAuth token (or GOOGLE_APPLICATION_CREDENTIALS fallback)
Bounce rate in context	High bounce interpreted by page type	GA4 runReport bounceRate (connected + data), else Manual	Connected (GA4)	OAuth token (or GOOGLE_APPLICATION_CREDENTIALS fallback)
Pogo-sticking risk	Users returning to SERP quickly = intent mismatch	GA4 + GSC derived (connected + GA4 data), else Manual	Connected (GA4+GSC)	OAuth token (or GOOGLE_APPLICATION_CREDENTIALS fallback)
Search intent alignment	Informational/navigational/commercial/transactional match	Gemini judges intent vs page type	Automated	GEMINI_API_KEY

Part 2 - GEO Parameters (fully automated)

01 Answerability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Concise top answer	Direct answer in first 40-50 words	Parse intro + Gemini judgement	Automated	Crawl + GEMINI_API_KEY
Question-phrased headings	H2/H3 as natural-language questions	Parse headings	Automated	Crawl (no key)
FAQ section	Marked up with FAQPage schema; long-tail variants	Parse JSON-LD + content	Automated	Crawl (no key)
TL;DR / summary box	Scannable summary easy for AI to lift	Detect summary + Gemini	Automated	Crawl + GEMINI_API_KEY

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Definition blocks (NEW)	"X is defined as..." near the top	Pattern + Gemini detection	Automated	Crawl + GEMINI_API_KEY
Step-by-step answers (NEW)	Numbered steps with clear action verbs	Parse ordered lists	Automated	Crawl (no key)
Comparison answers (NEW)	Tables or "A vs B" sections for comparative queries	Detect comparison tables	Automated	Crawl (no key)

02 Extractability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Semantic HTML	article/section/aside/figure used correctly; not div-soup	Parse DOM structure	Automated	Crawl (no key)
Chunkable paragraphs	Short paragraphs (2-4 sentences); one idea each	Analyze paragraph lengths	Automated	Crawl (no key)
Lists & tables	HTML ul/ol/table (not image-based)	Detect list/table elements	Automated	Crawl (no key)
Heading-to-content ratio (NEW)	Each heading followed by substantive content	Map headings to content blocks	Automated	Crawl (no key)
Code blocks / data blocks (NEW)	pre/code tags; machine-readable tables	Detect code/data markup	Automated	Crawl (no key)
No content in images/PDFs (NEW)	Text not locked inside images/PDFs	Detect text-in-image + Gemini OCR check	Automated	Crawl + GEMINI_API_KEY
Consistent terminology (NEW)	Same term used throughout; no synonym-switching	Gemini terminology analysis	Automated	GEMINI_API_KEY

03 Citability & E-E-A-T

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Author + credentials	Named author, bio, qualifications visible	Parse author block + Gemini	Automated	Crawl + GEMINI_API_KEY
Sources & citations	Outbound links to primary sources; references section	Parse citations/links	Automated	Crawl (no key)
Statistics & data	Quantitative claims with source attribution	Detect stats + Gemini	Automated	Crawl + GEMINI_API_KEY
Publish + updated dates	ISO 8601 dates in HTML + Article schema	Parse dates + JSON-LD	Automated	Crawl (no key)
About / Trust pages (NEW)	About, Contact, Privacy pages present	Fetch + detect trust pages	Automated	Crawl (no key)
Expert review signals (NEW)	"Medically reviewed by..."/"Fact-checked by..." labels	Pattern + Gemini detection	Automated	Crawl + GEMINI_API_KEY
Wikidata / Wikipedia entity (NEW)	Organisation/person entity exists	Query Wikidata + Wikipedia	Automated	Wikidata SPARQL + Wikipedia (no key)
Press mentions & awards (NEW)	Featured-in logos and award badges	On-page detection + Serper web search	Automated	Crawl + SERPER_API_KEY
Disclaimer / methodology (NEW)	Methodology explained for research/data content	Detect methodology + Gemini	Automated	Crawl + GEMINI_API_KEY

04 AI Structured Data

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
FAQPage schema	Q&A pairs in JSON-LD	Parse JSON-LD	Automated	Crawl (no key)
HowTo schema	Steps with name, text, image	Parse JSON-LD	Automated	Crawl (no key)
Article schema	headline, author, datePublished, dateModified, publisher	Parse JSON-LD	Automated	Crawl (no key)
Organization schema	name, url, logo, sameAs (socials, Wikidata)	Parse JSON-LD	Automated	Crawl (no key)
Product / Offer schema	name, description, price, availability, aggregateRating	Parse JSON-LD	Automated	Crawl (no key)
SpeakableSpecification (NEW)	Important passages marked for voice/AI	Parse JSON-LD	Automated	Crawl (no key)
Dataset schema (NEW)	For research/data pages	Parse JSON-LD	Automated	Crawl (no key)
Event schema (NEW)	For time-bound content	Parse JSON-LD	Automated	Crawl (no key)
ClaimReview schema (NEW)	Fact-check pages	Parse JSON-LD	Automated	Crawl (no key)

05 Entity Clarity

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
In-text definitions	First use of technical terms defined inline / linked	Detect definitions + Gemini	Automated	Crawl + GEMINI_API_KEY
Glossary page	Dedicated glossary with anchor links	Detect glossary page	Automated	Crawl (no key)
Named entities marked up	People/places/orgs in schema with sameAs to authoritative URIs	Parse JSON-LD sameAs	Automated	Crawl (no key)
Disambiguation (NEW)	Clarify ambiguous terms near first use	Gemini ambiguity analysis	Automated	GEMINI_API_KEY
Knowledge panel consistency (NEW)	Brand name/description/logo match Knowledge Panel	Query Google Knowledge Graph	Automated	GOOGLE_API_KEY
Entity co-occurrence (NEW)	Mention related authoritative entities	Gemini entity analysis	Automated	GEMINI_API_KEY

06 AI-Bot Crawlability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
GPTBot / ClaudeBot / PerplexityBot	Explicit allow/disallow in robots.txt	Parse robots.txt	Automated	Crawl (no key)
Google-Extended	Controls AI Overviews training data	Parse robots.txt	Automated	Crawl (no key)
llms.txt	Machine-readable site summary for LLMs	Fetch /llms.txt	Automated	Crawl (no key)
CCBot (Common Crawl) (NEW)	Explicit allow/disallow	Parse robots.txt	Automated	Crawl (no key)
Applebot-Extended (NEW)	Apple Intelligence crawler directive	Parse robots.txt	Automated	Crawl (no key)
Meta AI crawlers (meta-externalagent) (NEW)	Meta AI training/retrieval directive	Parse robots.txt	Automated	Crawl (no key)
Crawl rate limits (NEW)	AI bots not throttled to stale content	Parse Crawl-delay + measure response	Automated	Crawl (no key)
llms-full.txt (NEW)	Extended page-level metadata file	Fetch /llms-full.txt	Automated	Crawl (no key)

07 AI Quality Scores

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Clarity score	Writing unambiguous and scannable?	Gemini judgement	Automated	GEMINI_API_KEY
Authority score	Expertise, credentials, sourcing signaled?	Gemini judgement	Automated	GEMINI_API_KEY
Comprehensiveness	Covers topic fully enough to be self-sufficient?	Gemini judgement	Automated	GEMINI_API_KEY
Citation likelihood	Would an AI engine prefer this source?	Gemini judgement	Automated	GEMINI_API_KEY
Sample-query self-sufficiency test	Ask 5-10 likely queries; does page alone answer each?	Gemini multi-query test	Automated	GEMINI_API_KEY
AI Overview appearance monitoring (NEW)	Cited in Google AI Overviews / SearchGPT / Perplexity?	Run queries; check citations	Automated	SERPAPI_API_KEY + PERPLEXITY_API_KEY + OPENAI_API_KEY
Hallucination surface area (NEW)	Reduce ambiguous claims AI might misattribute	Gemini ambiguity/scope analysis	Automated	GEMINI_API_KEY
Prompt-align test (NEW)	Simulate LLM summary; accurate and favourable?	Gemini summary simulation	Automated	GEMINI_API_KEY
Competing source gap analysis (NEW)	Which sources AI cites for key queries that you lack	Diff citation lists vs target domain	Automated	SERPAPI_API_KEY + PERPLEXITY_API_KEY

08 Multimodal & Voice GEO

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Image alt text quality	Descriptive alts that match image content	Parse alts + Gemini vision check	Automated	Crawl + GEMINI_API_KEY
Video transcripts	Full text transcripts on-page or in schema	Detect transcript/VideoObject	Automated	Crawl (no key)
Conversational phrasing	Some content in natural spoken language for voice	Gemini tone analysis	Automated	GEMINI_API_KEY
Featured snippet shape	Match Google's voice/AI format: paragraph, list, or table	Detect content shape + Gemini	Automated	Crawl + GEMINI_API_KEY
SpeakableSpecification (voice)	Passages marked for text-to-speech	Parse JSON-LD	Automated	Crawl (no key)

Part 3 - GA4 & Search Console provisioning (connected mode)

The 6 first-party SEO parameters are measured when the audited domain is a **connected property** (listed in ``backend/connected_properties.json``) and Google credentials have access. GA4 and GSC only return data for properties the owner controls and has authorised; for non-connected domains these 6 fall back to Manual.

Roles: The **tool operator** creates one Google Cloud project (APIs, OAuth app, PageSpeed API key). The **site owner** owns GSC/GA4 and completes a one-time OAuth sign-in. See ``Connected_Mode_Tool_Operator_Guide.pdf`` and ``Connected_Mode_Site_Owner_Guide.pdf``.

Authentication (OAuth CLI - recommended)

Search Console often rejects service-account emails in the UI. The primary path is a **one-time browser sign-in** with the Google account that owns GSC/GA4:

File	Purpose
<code>`secrets/google-oauth-client.json`</code>	Desktop OAuth client JSON from Google Cloud Console
<code>`secrets/google-oauth-token.json`</code>	Refresh token saved by <code>`python scripts/google_auth.py`</code>
<code>`backend/connected_properties.json`</code>	Domain -> <code>{ gsc_site_url, ga4_property_id }</code> registry

Optional `.env`` overrides: ``GOOGLE_OAUTH_CLIENT_SECRETS``, ``GOOGLE_OAUTH_TOKEN_PATH``.

Google Cloud setup (tool operator's project - one project serves all connected sites):

1. Create a Google Cloud project (operator account).
2. Enable APIs: **Search Console API**, Google Analytics **Data** API, Google Analytics **Admin** API (Admin API lists GA4 Property IDs in the auth script), **PageSpeed Insights API** (for Core Web Vitals; uses ``GOOGLE_API_KEY``, not OAuth).
3. OAuth consent screen: External; scopes ``https://www.googleapis.com/auth/webmasters.readonly`` and ``https://www.googleapis.com/auth/analytics.readonly``. Add test users while app is in Testing mode.
4. Credentials -> OAuth client ID -> **Desktop app** -> download JSON -> ``secrets/google-oauth-client.json``.
5. Credentials -> **API key** -> restrict to PageSpeed Insights API -> set ``GOOGLE_API_KEY`` in `.env``.

Run the auth script (from the ``backend`` folder):

```
``bash
cd backend && source .venv/bin/activate
python scripts/google_auth.py          # one-time browser sign-in
python scripts/google_auth.py --list-only # re-list GSC/GA4 without browser
``
```

If your shell is already in ``backend``, omit the ``cd backend &&`` prefix.

The script prints GSC site URLs and numeric GA4 Property IDs. Copy the exact values into ``connected_properties.json``, restart the backend, and re-run an audit.

Credential priority at audit time: OAuth token file (``secrets/google-oauth-token.json``) -> service account JSON (``GOOGLE_APPLICATION_CREDENTIALS``). OAuth wins when both exist.

Adding another website: Do not create a new Cloud project. Add a row to ``connected_properties.json`` and sign in with a Google account that has access to that site's GSC/GA4 (re-run ``google_auth.py`` if the owner account differs).

Authentication (service account - optional / agency)

When GSC accepts a service-account email, set ``GOOGLE_APPLICATION_CREDENTIALS`` to the JSON key path. The site owner adds that email in GSC (user) and GA4 (Viewer). Same ``connected_properties.json`` registry applies. OAuth takes priority when ``google-oauth-token.json`` exists. Not covered in the Connected Mode setup PDFs (OAuth-only).

What is measured once GA4 + GSC are connected

Parameter (was Manual when not connected)	Source	API method	Data returned
Click-through rate (CTR)	GSC	searchanalytics.query	clicks, impressions, ctr, average position (per page); **Not Meeting** when connected but 0 impressions
Sitemap submission status	GSC	sitemaps.list	submitted paths, errors, warnings; **Not Meeting** when 0 sitemaps, **Partial** when errors
Log file analysis (crawl frequency)	GSC	urlInspection.index.inspect	lastCrawlTime, coverageState, verdict, pageFetchState, robotsTxtState; **Partial** when crawled but not indexed
Dwell time / scroll depth	GA4	runReport (28daysAgo-yesterday, pagePath filter)	averageSessionDuration, bounceRate, engagementRate (per pagePath; **engagementRate fetched in a separate request** - incompatible with bounceRate in one GA4 call)
Bounce rate	GA4	runReport	bounceRate (per page)
Pogo-sticking risk	GA4 + GSC	runReport + derived	engagementRate when GA4 data exists

Performance Baseline PDF analytics (Connected Mode)

When `kind=performance\_baseline` and the domain is connected, `Ga4Client.fetch\_baseline()` pulls a 90-day property-level bundle in parallel with GSC dimensions:

Baseline PDF metric	GA4 / GSC source
Engagement Rate, Total Users, New Users, Bounce Rate, Avg Session Duration, Pages per session	GA4 property overview (+ separate engagementRate query)
Clicks by device / country, organic traffic trend, top keywords	GSC searchAnalytics
Traffic by channel, session sources, Organic Social, device split, bounce/engagement trends, top pages, forms	GA4 runReport dimensions

Resolution order for ****Engagement Rate**** in the baseline PDF: property overview -> page-level engagement -> Pogo-sticking risk parameter detail.

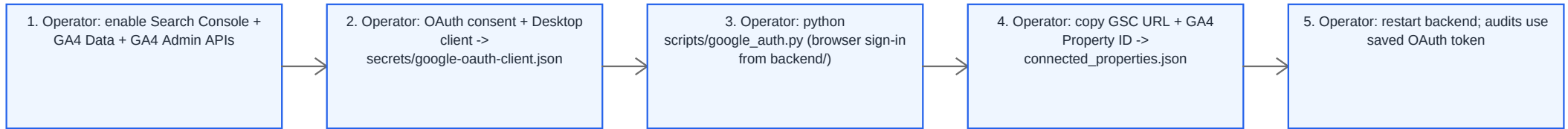
First-party rating outcomes (connected mode)

Outcome	Rating	Counts in score?	When
API success + pass/fail logic	Meeting / Partial / Not Meeting	Yes	Normal path (e.g. CTR with impressions, sitemap submitted cleanly, URL indexed and crawled)
Sitemap: 0 registered in GSC	Not Meeting	Yes	Connected, sitemaps.list succeeded
Sitemap: submitted with GSC errors	Partial	Yes	Connected, at least one sitemap path has errors
CTR: 0 GSC impressions for page	Not Meeting	Yes	Connected, searchanalytics succeeded but no rows
Log file: crawled, not indexed	Partial	Yes	URL Inspection returned crawl data but verdict/coverage not indexed
URL Inspection empty	Not Measured	No	Connected but Google has not crawled/indexed the URL yet
API call failed after retries	Not Measured	No	GSC/GA4 network or auth error (`first_party_empty`)
Domain not connected	Manual	No	Missing from `connected_properties.json` or no credentials
GA4 connected but no sessions	Manual	No	GA4 engagement rows until runReport returns a row for the page path

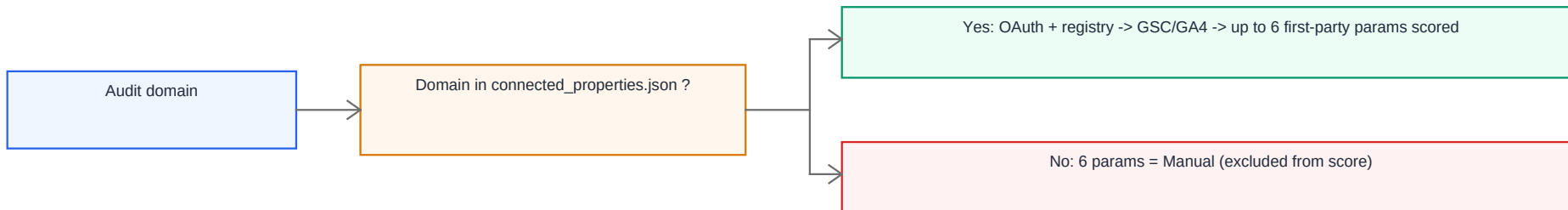
What stays Manual even with GA4 + GSC

Parameter	Why it stays manual
Backlink profile	GSC Links report has no API
Referring domain count	GSC Links report has no API
Competitor link gap	Needs off-site link index; competitor data never in your GSC
Brand mentions (unlinked)	Not exposed by GA4/GSC
Google Business Profile / NAP cross-directory	Needs Business Profile / Places + directory data

Provisioning flow (one-time setup)



How the tool decides per audit



Connected-property registry

The orchestrator reads `backend/connected\_properties.json` at audit time. Host is normalised (lowercase, `www.` stripped). Example:

```
```json
{
 "example.com": {
 "gsc_site_url": "https://www.example.com/",
 "ga4_property_id": "123456789"
 }
}
```

`gsc\_site\_url` must match the GSC property URL exactly (including trailing slash). `ga4\_property\_id` is the numeric Property ID from GA4 Admin -> Property settings (not the `G-XXXXXXXX` Measurement ID).

For a multi-tenant product this mapping is collected via an onboarding form and stored in a database; the lookup logic is identical.

## GSC client behaviour (`GscClient` in `app/clients.py`)

- All GSC calls return a structured `GscApiResponse` (`ok`, `data`, `error`) instead of raising to agents.
- **Retries:** up to 3 attempts with backoff (0.5s, 1s, 2s) on transient failures; cached API client is reset between attempts.
- **Concurrency:** an `asyncio.Lock` serialises GSC calls within one audit (sitemap, CTR, URL Inspection) because the Google API client is not safe for parallel `execute()` on a shared connection.
- **URL variants:** Search Analytics and URL Inspection try common URL forms (with/without trailing slash, www vs non-www) before returning empty data.
- **Search Analytics:** date range 2020-2030, `dataState=final`, page dimension filter.

## Implementation modules

Module	Role
`app/google_auth.py`	Load OAuth token or service account; refresh expired tokens
`scripts/google_auth.py`	One-time CLI sign-in; `--list-only` to re-list properties
`app/clients.py` (`GscClient`, `Ga4Client`, `GscApiResponse`)	GSC sitemaps / searchanalytics / urlInspection; GA4 runReport + `fetch_baseline()`; PSI via `GOOGLE_API_KEY`
`app/baseline_analytics.py`	`fetch_baseline_analytics()` - parallel GA4/GSC bundle for baseline PDF
`app/agents/seo_agents.py`	`_sitemap_gsc`, `_log_analysis`, `_ctr_row` - connected-mode First-party logic
`app/agents/site_audit_supplement_agent.py`	Shallow crawl supplement: issue counts, site health, link distribution
`app/agents/base.py`	`scored`, `manual`, `first_party_empty`, `not_measured` result factories
`app/orchestrator.py` (`ConnectedRegistry`)	Domain lookup at audit time; progress callbacks
`app/reference_pdf_catalog.py`	Parameter catalogs for three supplementary PDFs
`app/reference_pdf_export.py`	PDF builders for baseline, full site audit, AI overview
`app/reference_pdf_layout.py`	ReportLab layout primitives, score badges, tables
`app/audit_history.py`	SQLite store + trend chart data for Full Site Audit PDF
`app/ai_health.py`	AI Search Health composite score
`backend/connected_properties.json`	Static registry (JSON file today)
`scripts/build_pdf_parameter_comparison.py`	Semrush vs generated PDF comparison report
`scripts/run-dev.sh` / root `package.json`	Single-command dev server
`generate_connected_guide_pdf.py`	Professional PDFs for site-owner and tool-operator setup guides

## Limits and behaviour

Topic	Detail
Data latency (GSC)	Search performance data has a 2-3 day delay; the tool requests `dataState=final`
CTR with 0 impressions	Connected + GSC API success but no rows for the page -> <b>**Not Meeting**</b> (scored), with recommendation to improve indexing/traffic
GSC/GA4 API failure	After retries, parameter is <b>**Not Measured**</b> (`first_party_empty`) with error detail in the report
Empty URL Inspection	Connected but no inspection result -> <b>**Not Measured**</b> until Googlebot crawls the URL
GA4 without sessions	Dwell / bounce / pogo rows stay <b>**Manual**</b> until GA4 returns engagement data
GA4 engagementRate + bounceRate	Must be requested in <b>**separate**</b> `runReport` calls (GA4 metric incompatibility); merged in client
Audit history DB	`backend/data/audit_history.db` (gitignored); trend charts appear after >=2 runs with changed metrics
PageSpeed Insights	Requires PageSpeed Insights API enabled in the tool operator's Cloud project and `GOOGLE_API_KEY` in `.env`; otherwise CWV/PSI parameters are <b>**Not Measured**</b>
URL Inspection quota	2,000 inspections per property per day; the tool inspects the audited URL only
Privacy	Read-only scopes (`webmasters.readonly` + `analytics.readonly`); owner can revoke OAuth access or remove users at any time
Fallback	If a domain is not in the registry (or credentials missing), the 6 first-party parameters revert to Manual and are excluded from the score

## Part 4 - APIs and keys summary

### HTTP API (selected)

Method	Path	Purpose
GET	`/api/health`	Health check
POST	`/api/audit`	Run full audit; returns SEO + GEO JSON reports
POST	`/api/audit/stream`	Same audit with SSE progress events
POST	`/api/report/pdf`	Download SEO or GEO report PDF
POST	`/api/report/supplementary-pdf`	Download baseline / full site audit / AI overview PDF

Capability	API / source	.env key	Status
AI judgement (SEO + GEO)	Gemini (google-gen ai)	GEMINI_API_KEY	Present
ChatGPT-style citation check	OpenAI	OPENAI_API_KEY	Present
Performance + entity	PageSpeed Insights + Knowledge Graph	GOOGLE_API_KEY (enable PageSpeed Insights API in tool operator GCP project)	Required for CWV/PSI rows
Web search + press mentions	Serper	SERPER_API_KEY	Present
Perplexity citations	Perplexity Sonar	PERPLEXITY_API_KEY	Present
Google AI Overview citations	SerpApi	SERPAPI_API_KEY	Present
Crawl/render + extraction	Firecrawl	FIRECRAWL_API_KEY	Present
Brand mentions on X	X API	X_BEARER_TOKEN	Present
First-party analytics (GA4 + GSC)	Search Console + GA4 Data API (OAuth CLI primary)	`secrets/google-oauth-token.json` (+ optional `GOOGLE_APPLICATION_CREDENTIALS` fallback)	Per connected domain
Entity graph	Wikidata SPARQL + Wikipedia REST	none	No key needed
PDF generation	ReportLab + Matplotlib	none (Python deps in `requirements.txt`)	Built-in

All other API keys are read from `.env` at startup via `app/config.py`. PageSpeed and first-party Google access are independent: PSI uses `GOOGLE\_API\_KEY`; GSC/GA4 use OAuth (or service-account fallback).

## Part 5 - Supplementary PDF reports (detail)

Three client-facing PDFs mirror Semrush export layouts. Each uses `reference\_pdf\_catalog.py` to map external metric names to existing auditor parameters (or documents why a metric is not calculated).

### 1. Performance Baseline (`performance\_baseline`)

- \*\*23 parameters:\*\* GA4 overview KPIs (engagement, users, bounce, session duration), GSC device/country/keywords/pages, GA4 channel/source/social/device reports, CWV (LCP/CLS/INP), load speed, form submissions, exit pages, bounce/engagement trend charts.

- **Connected Mode required** for most traffic metrics; CWV from PSI regardless.
- **Layout:** cover -> executive summary (SEO score badge) -> Overview KPI grid (bordered) -> sectional tables and Matplotlib charts.
- **Analytics entry point:** `fetch_baseline_analytics()` called from `POST /api/report/supplementary-pdf` when `connected=true`.

## 2. Full Site Audit (`site_audit_full`)

- **99 parameters:** Semrush-style errors, warnings, notices from shallow crawl supplement agent; site health (= SEO score); crawled page status; internal link distribution; optional trend charts from `audit_history.db`.
- **Site audit supplement agent** runs after primary SEO crawl; populates issue inventory and link buckets.
- **Executive summary:** single Site Health badge + error/warning/notice counts; issue tables use severity colour only when count > 0.

## 3. AI Search Overview (`ai_search_overview`)

- **12 parameters:** AI Search Health score (`ai_health.py`), aggregate AI bot blocking %, per-bot robots.txt rules (ChatGPT-User, OAI-SearchBot, Google-Extended, Perplexity, Claude, etc.), crawl status breakdown, top issues list.
- **Site Health** intentionally excluded (reported in Full Site Audit instead).
- **Executive summary:** AI Search Health badge only + blocked-pages percentage + actionable GEO/SEO issue bullets (dashboard/meta parameters filtered from action items).

## Parameter resolution

For each catalog row, `reference_pdf_export.py`:

1. Resolves from SEO/GEO `ParameterResult` index via `app_keys`.
2. Overrides with live GA4/GSC data when `BaselineAnalytics` is available (baseline PDF).
3. Marks **Manual** with catalog `manual_reason` when data cannot be obtained.

Manual-parameter appendix sections were removed from client PDFs; reasons appear only in the internal comparison PDF and catalog.

## Audit history and trends

`audit_history.py` stores per-domain snapshots (site health, crawled pages, error/warning/notice counts) in SQLite on each audit when metrics change. Full Site Audit PDF embeds 2x2 mini trend charts when >=2 distinct snapshots exist for the domain.